

Startup Team Charter

Team Information

Startup Name: CoursePilot

Course: CSCI 577a – Software Engineering

Term: Spring 2026

Team Members (Name, Role, Email):

- Michael Nguyen, Product Manager, nguyenmq@usc.edu
- Matt Zandi, Backend Engineer, mczandi@usc.edu
- Evan Tomaine, Backend Engineer, tomaine@usc.edu
- Pradeep Mathiyarasu, AI Engineer, mathiyar@usc.edu
- Weston Trautmann, Backend Engineer, wtrautma@usc.edu
- Daniel Amoako-Frimpong, Frontend Engineer, amoakofr@usc.edu
- Joshua Oti, AI Engineer, aoti@usc.edu
- Lauryn Do-Diallo, Frontend Engineer, dodiallo@usc.edu
- Nicolette Fulcomer, Frontend Engineer, nfulcome@usc.edu
- Vaidik Vadhavana, Backend Engineer, vadhavan@usc.edu

Mission & Vision

Mission Statement:

Our mission is to design and build an AI-powered student dashboard that helps busy college students organize academic responsibilities by transforming syllabi into actionable schedules, reminders, and study support tools, enabling more effective time management and reduced cognitive load.

Vision Statement:

A future where students always know what's next and rarely fall behind, because their academic lives are organized for them instead of by them. **CoursePilot** will become an intelligent, centralized AI command center that continuously ingests real course workloads, surfaces what matters most right now, and empowers students to stay on track with minimal effort.

Problem Statement & Value Proposition

Problem Being Solved:

Students suffer from "Digital Sprawl." Their deadlines are in a PDF syllabus, their lecture slides are on Brightspace or Canvas, their personal notes are in Google Docs, and their flashcards are in a separate app. Constantly switching between these contexts causes students with demanding schedules to have missed deadlines, inefficient time management, and increased cognitive overload. The problem is the lack of a centralized, low-effort system that converts course requirements into actionable, timely guidance.

Target Users / Customers:

Busy students, including full-time students balancing work, extracurriculars, or multiple courses, who struggle with organization and efficient study habits.

Unique Value Proposition:

The platform will feature a comprehensive dashboard that offers full-stack customizable academic automation:

1. **Universal Ingestion:** Upload Syllabi, PPT Slides, and Document Notes. AI will extract important dates and upload them into their calendar, push notifications, and send reminders along with key concepts for studying.
2. **Generative Study Aids:** The system proactively converts uploaded materials into study guides and summaries. It offers assignment complexity estimation and can give study session recommendations depending on the content of the course material.
3. **Ecosystem Integration:** Unlike standalone dashboards, the platform will be integrated with public APIs (i.e. calendar and flashcard tools) to sync the student's plan with their existing digital life.

Team Roles & Responsibilities

Describe each team member's role and primary responsibilities:

- **Product Manager:** Owns the product vision and manages the MVP Scope to ensure we ship a polished, working product. Creates user stories and takes the lead of designing the system's microservices architecture.
- **Frontend Engineer:** Leads the implementation of the UI code. Translates UI/UX Designer files into pixel-perfect code, focusing on smooth transitions and a responsive layout.
- **Backend Engineer:** Builds and maintains the server side of the app. Focus on the "behind the scenes" logic of the app, including databases, APIs, and server infrastructure to power features that users will not see directly.
- **AI Engineer:** Design and implement AI-driven document understanding pipelines for educational content extraction and normalization; develop intelligent scheduling and prioritization algorithms using LLM-based reasoning.

Decision-Making & Governance

Decision-Making Model (e.g., consensus, majority vote, role-based):

1. Technical Decisions (Role-Based):

Technical implementation details are delegated to the experts in that domain. For example, Backend Engineers will decide the database schema, Frontend Engineers decide the state management library, and AI Engineers can select the specific LLM

models used. If a disagreement arises within a role group, the rest of the team will facilitate a discussion, but the final decision rests with the corresponding team members.

2. Product Decisions (Decision Matrix):

Decisions regarding feature scope, roadmap prioritization, or user flows will use a Decision Matrix. Team members will rate competing options on a scale of 0-5 based on key criteria (i.e. user value, ease of implementation, risk). The scores will be summed, and the option with the highest total will be selected. This ensures objectivity in product strategy.

3. General Team Decisions (Majority Vote):

For general governance, scheduling, or high-level conflict resolution, the team will utilize a majority vote.

Conflict Resolution Approach:

Interpersonal or professional conflicts will be addressed first through direct, private dialogue between the involved parties. If the issue remains unresolved, it will be escalated to the Product Manager for mediation. If the conflict persists and impacts project progress, it will be brought to the entire team during a retrospective for a group resolution with final arbitration by a course instructor if necessary.

Communication Plan

Primary Communication Tools (e.g., Slack, Teams, Email):

Slack will serve as our primary platform for all day-to-day communication. We will use email only for formal communication with external stakeholders or instructors. Code reviews and technical discussions will take place directly within Github pull requests and during meetings as necessary. The text group chat will only be used when there is a need for critical, time-sensitive updates or immediate action items.

Meeting Cadence (standups, sprint planning, retrospectives):

The team will hold a mandatory asynchronous meeting every Tuesday at 8:30 PM. This time block will be utilized for our weekly standup to align on immediate goals, followed by Sprint Planning or Retrospectives as dictated by the sprint schedule. Ad-hoc technical syncs may be scheduled by sub-teams as needed, but the Tuesday night slot will be used for full-team coordination.

Development & Quality Practices

Development Methodology (e.g., Agile/Scrum):

We will adhere to the Agile Scrum methodology, operating in two-week sprints. This

iterative approach allows us to rapidly prototype, test, and refine features based on feedback.

Definition of Done:

A feature is considered “Done” only when the code has been written, met all the acceptance criteria, passed all automated unit tests, received approval via peer review, been merged into the main branch, and successfully deployed without any regression errors.

Quality Metrics & Review Practices:

The team will maintain high code quality by requiring at least one peer review for every pull request. We will track metrics such as bug density, test coverage (aim for higher than 80%), and sprint velocity.

Commitment & Expectations

Expected Weekly Time Commitment per Member:

Each team member will commit to contributing approximately 8-10 hours per week on the project. This includes time spent in meetings, individual development work, documentation, and communication.

Availability Constraints:

We acknowledge that members have other academic and professional obligations. Provided that sprint goals are on track, we will not expect immediate responses during “quiet hours” overnight or on weekends. All members are expected to attend the Tuesday 8:30 PM meeting unless an absence notice is provided beforehand.

To proactively manage sprint velocity, team members have identified the following dates where their availability will be significantly reduced (i.e. midterms, travel, etc.).

Known Busy Periods	
Team Member Name	Dates Unavailable
Michael Nguyen	2/18 - 2/20, 4/1 - 4/3, 4/9 - 4/12
Matt Zandi	-
Evan Tomaine	2/14 - 2/16
Pradeep Mathiyarasu	-
Weston Trautmann	1/30-2/1

Daniel Amoako-Frimpong	-
Joshua Oti	-
Lauryn Do-Diallo	3/12-3/16
Nikki Fulcomer	1/23-1/25
Vaidik Vadhavana	-

Risk Management

Key Risks (technical, schedule, team):

Technical:

1. **AI Hallucinations:** Generative AI may hallucinate facts when summarizing lecture slides (i.e. inventing a new definition that sounds plausible but is factually incorrect) which could cause a student to fail an exam they are studying for.
2. **Unstructured PDF Parsing:** Documents such as Syllabi can vary wildly in format (involving tables, columns, images). Standard Optical Character Recognition (OCR) may incorrectly associate dates with assignments.

Schedule:

1. **Integration Delay:** The Frontend and Backend teams might work in isolation for too long which can lead to massive breakage (i.e. connecting a React Dashboard to an API in the final weeks).
2. **Midterm Availability:** Multiple team members have exams around the same week which could potentially drop sprint velocity.

Team:

1. **Skill Silos:** Specific knowledge (i.e. AI pipelines or Database Schema) might be held by one person. If they are busy or unavailable, work stops.
2. **Communication Overhead:** In a team of 10, decision-making can become slow, and messages can get lost in a noisy Slack channel.

Mitigation Strategies:

Technical:

1. **AI Hallucinations:** The team will use Retrieval Augmented Generation (RAG) with strict citations. The AI must be prompted to only use information present in the source text and display a source link (page number) next to every generated summary so users can verify accuracy.

2. **Unstructured PDF Parsing:** Implement a "Human-in-the-Loop" validation step. After parsing, the user is presented with the extracted dates and must confirm or edit them before they are finalized in the calendar.

Schedule:

1. **Integration Delay:** Implement an API-First workflow. The Backend team must deliver an API specification in Week 1. The Frontend team will develop against this contract using mock data, ensuring continuous integration throughout the semester.
2. **Midterm Availability:** We have mapped blackout dates above. The Product Manager will schedule "lighter" maintenance sprints during these heavy academic weeks and front-load complex features into clearer weeks.

Team:

1. **Skill Silos:** Enforce Pair Programming on critical tasks and require that Pull Requests are reviewed by someone outside the immediate sub-team occasionally to spread system knowledge.
2. **Communication Overhead:** We will strictly use threaded conversations in Slack and require that all decisions made in chat are documented in the project Jira/Wiki. We will rely on the Role-Based decision model to prevent 10-person debates on minor technical issues.

Ethics & Academic Integrity

Approach to ethical decision-making:

We will prioritize data privacy, ensuring user files are encrypted. The AI will be tuned to provide study support (quick summaries, explanations) rather than academic dishonesty (per syllabus requirements).

AI usage and disclosure plan (per syllabus requirements):

The team will utilize Generative AI tools (i.e. Gemini, Claude, ChatGPT) to assist with coding and debugging, acting as a "pair programmer." However, we will explicitly disclose this usage in the project documentation. We will not use AI to generate the core intellectual content of our reports or to bypass the fundamental learning objectives of the course. All AI-generated code will be manually reviewed, understood, and tested by the team.

Amendments & Review

Process for updating this charter:

Any team member can propose an amendment during our regular Tuesday meetings. Proposed changes will be discussed openly, and amendments require a majority vote to pass.

Signatures

By signing below, team members agree to adhere to this charter:

Name / Signature / Date

Michael Nguyen 1/ 25/ 2026

Matt Zandi 1/ 25/ 2026

Evan Tomaine 1/ 25/ 2026

Pradeep Mathiyarasu 1/ 25/ 2026

Weston Trautmann 1/ 25/ 2026

Daniel Amoako-Frimpong 1/ 25/ 2026

Joshua Oti 1/ 25/ 2026

Lauryn Do-Diallo 1/ 25/ 2026

Nikki Fulcomer 1/ 25/ 2026

Vaidik Vadhavana 1/ 25/ 2026